Virenpuri Vasantpuri Vairagi

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LinkedIn | GitHub | Portfolio | Live ML App

Professional Summary

Aspiring Python Developer and Data Scientist with hands-on experience designing automated machine learning pipelines and scalable web applications. Highly proficient in Python Fundamentals, and Data Structures. Demonstrated ability to manage the full software development lifecycle, from raw data preprocessing to deploying real-time applications via Streamlit. Skilled to build intelligent, data-driven systems that enhance business efficiency and solve complex operational challenges.

Education

S.P.B Patel Engineering College — Mehsana, Gujarat
Bachelor of Engineering in Computer Engineering (CGPA: 8.00) — 2022 – 2026

Experience

Brainy Beam Info-Tech Pvt. Ltd. — Ahmedabad, Gujarat
Data Science & Machine Learning Intern — 01/2026 – 03/2026

- Architected the complete end-to-end Data Science pipeline to develop a highly accurate job offer acceptance prediction system, drastically improving human resources forecasting capabilities.
- Processed and encoded target variables, specifically mapping 'Joined' to 1 and 'Not Joined' to 0, ensuring precise model evaluation and pristine training data integrity.
- Evaluated multiple classification algorithms comprehensively, conclusively determining that Random Forest outperformed alternatives such as Gradient Boosting in precision and reliability.
- Deployed the finalized machine learning model to a live Streamlit Cloud environment, empowering recruitment teams to conduct immediate, real-time applicant analysis.

InfoLabz IT Services Pvt. Ltd. — Ahmedabad, Gujarat
Data Analysis & Machine Learning Intern — 07/2025 – 07/2025

- Constructed clean, automated data pipelines utilizing Python, Pandas, and Scikit-learn to streamline extensive exploratory data analysis (EDA).
- Executed rigorous data preprocessing protocols that significantly enhanced the readiness, accuracy, and reliability of datasets for supervised learning evaluations.

Projects

Job Offer Acceptance Prediction
Technologies: Python, Pandas, Scikit-Learn, Random Forest, Streamlit

- Formulated an advanced machine learning solution to forecast candidate dropouts, equipping technical recruiters with actionable, data-driven predictive insights.

- Refined raw salary expectation data through meticulous cleaning and EDA, extracting critical behavioral indicators that strengthened the model's overall predictive power.
- Engineered and launched an interactive web application via Streamlit Cloud, utilizing the optimized Random Forest algorithm to deliver seamless, real-time forecasting results to end-users.

AI-Powered Customer Feedback Classification
Technologies: n8n, Google Gemini, Airtable, APIs

- Built a production-ready automation workflow combining n8n and Google Gemini to seamlessly capture, parse, and analyze unstructured customer feedback.
- Categorized incoming data streams dynamically into actionable segments (Complaints, Compliments, Feature Requests), significantly reducing manual review time.
- Orchestrated continuous database synchronization and automated team notifications utilizing Airtable and Slack APIs to accelerate internal response times.

Adaptive & Collaborative Traffic Control System
Technologies: AI/IoT, Random Forest, V2X, Streamlit

- Integrated real-time IoT sensor data with machine learning frameworks to accurately monitor traffic density and predict urban congestion patterns.
- Designed a collaborative decision-making platform leveraging V2X communication to automate emergency vehicle prioritization, ensuring highly efficient path-clearing for ambulances.
- Implemented an adaptive traffic signal control mechanism by feeding continuous IoT data streams directly into a Random Forest classifier to achieve dynamic system responsiveness.

Publications & Extracurricular

- **Publication:** Published peer-reviewed research titled "Multi-feature Collaborative Traffic Management: An AI and IoT-Integrated Approach" in the 5th International Conference on Advanced Computational and Communication Paradigms, Springer (DOI: 10.1007/978-3-032-13544-5_39), February 2026.
- **Hackathon:** Awarded 3rd Prize at the Smart India Hackathon (SIH) 2025 out of numerous competitive entries by prototyping an innovative Smart Traffic Management System aimed at mitigating urban congestion.

Technical Skills

- **Languages & Databases:** Python, SQL, HTML5, CSS3, JavaScript, PHP, Java, C
- **Frameworks & Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Streamlit
- **AI & Automation:** LLM Integration, n8n, Model Context Protocol (MCP)
- **Software & Tools:** n8n, PyCharm, Jupyter Notebook, VS Code, Power BI, Airtable, Tableau, GitHub/Git, XAMPP